

Shaojie Li

Date of Birth: 1999.06.26

Research Interests: Embodied Intelligence / Multimodal Agents /
Reinforcement Learning / Computer Vision /
Robotics / XR/AR Reality Grounding

Research Assistant at The Chinese University of Hong Kong. Interests:
embodied intelligence, multimodal agents, reinforcement learning,
computer vision, robotics, and uncertainty-aware grounding for XR/AR
systems.

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Education

◆ 2023.05 - 2024.08 **University of New South Wales** Artificial Intelligence Master

Relevant coursework: artificial intelligence, machine vision, algorithm design, neural networks, graph theory, AI project design, game theory, database systems, and front-end engineering.

◆ 2017.06 - 2021.06 **Shenzhen University** Internet of Things Engineering Bachelor

Relevant coursework: software engineering, computer networks, data structures, database systems, digital electronics, microcomputer principles and interfaces, operating systems, and information security.

Publications & Invention Patents

- [1] Replay-time Closed-loop Evaluation Protocol for Learned 3D Surgical Dynamics / World Models[C]. NeurIPS 2026 Evaluations & Datasets Track, 2026. Submitted / Under Review. **Second author**

Research Projects

◆ RoboStore Unmanned Retail Terminal System

(Intelligent Retail System / Integrated Hardware-Software Platform) 2025.05.21 - Present

Role: Responsible for robotic arm control, beverage recognition with computer vision, and device integration.

Developed an end-to-end unmanned retail terminal system covering vision-assisted recognition, robotic-arm dispensing, order payment, inventory management, refund handling, and reconciliation workflows.

#Computer Vision #Robotic Arm Control #FastAPI #SQLAlchemy #Device Integration

◆ Prostate Biopsy Robot (Medical Robot / Ultrasound-guided Puncture System)

2025.08.15 - Present

Role: Responsible for control of the collaborative robotic arm and prostate puncture mechanism, as well as device integration.

Built a robotic biopsy workflow in which the doctor positions the robotic arm, after which the system performs automatic puncture while real-time ultrasound displays the needle location.

#Collaborative Robot #Medical Robot #Real-time Ultrasound #Vision Guidance

◆ Intelligent Single-scale Flow Evaluation System (Medical Health / Time-series Signal Analysis)

2026.03.02 - Present

Role: Responsible for weight time-series data cleaning, segmentation, manual window annotation tooling, dataset construction, 1D CNN model training, and single-file inference UI development.

Predicted fluid flow curves, cumulative volume curves, and core evaluation metrics from a single high-precision weight sensor; upgraded the model from direct cumulative-curve prediction to a physics-constrained flow-first structure followed by integration.

#Time Series #1D CNN #NumPy #Pandas #Model Evaluation

◆ Monocular Projection 3D Imaging Platform

(Research Platform Development / Computer Vision) 2021.11 - 2023.04

Role: Responsible for interaction logic and serial communication support for the camera, projector, and point-cloud display modules.

Developed a research and demonstration platform for 3D imaging, focusing on coordinated visual-device control and imaging workflow implementation.

#Qt #Serial Communication #3D Imaging #Research Platform

◆ Defect Detection Platform (Intelligent Inspection Platform / Engineering Implementation)

2021.11 - 2023.04

Role: Responsible for Python module execution, system interaction logic, and database management.

Built a platform for defect detection scenarios, supporting inspection workflows, data flow, and functional module invocation.

◆ **HalfCheetah Reinforcement Learning Project** (Reinforcement Learning) Master's Project

Role: Responsible for tuning the TQC model and modifying the reward mechanism to improve the locomotion performance of the HalfCheetah robot.

Trained a neural-network-based reinforcement learning agent to maximize locomotion distance in a 2D environment, focusing on model tuning and reward design.

#Reinforcement Learning #Neural Networks #TQC

◆ **Jigsaw Toxic Comment Severity Rating** (Deep Learning / Natural Language Processing)

Master's Project

Role: Responsible for training and comparing multiple deep learning models, including GPT-3.5, GPT-2, BERT, and GRU.

Built comparative experiments on the Kaggle toxic comment dataset to rate comment toxicity across multiple dimensions.

#NLP #BERT #GPT #GRU

◆ **Turtle and Penguin Classification** (Supervised Learning / Computer Vision) Master's Project

Role: Responsible for image preprocessing, HOG feature extraction, and SVM model training.

Implemented automatic image recognition and processing in Python to train and evaluate turtle and penguin classification models.

#Computer Vision #HOG #SVM #Python

◆ **Matlab-based Image Processing Program** (Image Processing) Undergraduate / Master's Project

Role: Responsible for automatic image recognition and processing, including object recognition, counting, and segmentation.

Implemented coin denomination and quantity recognition, as well as automatic person recognition and individual segmentation in group photos.

#Matlab #Image Segmentation #Object Recognition

◆ **Remote Unattended Terminal Network Fault Solution** (Embedded System Development)

Undergraduate Project

Role: Developed C programs for ESP-32 on Arduino, enabling dual-mode network and GSM SMS communication control.

Implemented remote control of ESP-32 power switching and lighting, combining embedded systems with communication solutions.

#Arduino #ESP-32 #Embedded Systems #GSM

Relevant Experience

- ◆ **The Chinese University of Hong Kong (CUHK)** Research Assistant 2025.05.21 - Present
- ◆ **Hong Kong Centre for Logistics Robotics** Assistant Engineer 2021.11 - 2023.04

Technical Skills

- ◆ Research methods: reinforcement learning, computer vision, deep learning, embodied intelligence
- ◆ Systems and experiments: machine learning modeling, image processing, model training and evaluation, end-to-end intelligent system implementation
- ◆ Platforms and devices: robotic system integration, embedded development, serial communication, device control and terminal deployment
- ◆ Programming and tools: Python, C++, C, MATLAB, JavaScript, HTML/CSS, FastAPI, Qt, MySQL
- ◆ English proficiency: IELTS 7.0